CURRICULUM VITAE

Jinho Byun

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RESEARCH EXPERIENCE

Mar. 2025 – Present	Research professor at Dept. of Energy Engineering, KENTECH
Mar. 2023 – Mar. 2025	Post-doctoral researcher at Dept. of Energy Engineering, KENTECH
Aug. 2022 – Jan. 2023	R&D engineer at Wonik IPS
Mar. 2022 – July 2022	Post-doctoral researcher at Dept. of Physics, Pusan National Univ. Advisor: Prof. Jaekwang Lee

EDUCATION

Sep. 2017 – Feb. 2022	Pusan National University, Ph.D. Major: Physics Advisor: Prof. Jaekwang Lee Thesis: Emergent Phenomena at Polar Oxides and Their Applications
2012 - 2017	Pusan National University, B.A. Major: Physics

HONORS AND AWARDS

2021 **Outstanding Poster Award**
Awarded by the Korean Physics Society (KPS) for Poster entitled: "Interface engineered TER optimization in HfO_2-based FTJ"

2021 **Excellent Academic Award**
Awarded by Pusan National University.

2017-2020 **Combined course Scholarship**
Full amount of school support fees funded by Pusan National University.

2020 **Humantech Paper Award**

Awarded by Samsung Electronics for the paper entitled: "Anisotropic Evaporation of ZnO Observed by In-situ Cs-Corrected High Resolution TEM"

2019 **Outstanding Poster Award**
Awarded by Korea Institute for Advanced Study (KIAS) for Poster entitled: "First-principles study of the HfO_2-based ferroelectric tunnel junction"

2017 **Outstanding Presentation award**
Awarded by the Korean Physics Society (KPS) for presentation entitled: "Design of high-coercivity $Fe_{16-x}Al_2N_2$ alloy".

2012-2017 **National Excellence Scholarship (Natural Sciences and Engineering)**
Full amount of school support fees funded by Korean Government.

PUBLICATIONS

1. Y. Xing¶, **J. Byun¶**, J. Seo, Z. Wang, Y. Jin, S. Kim, J. Lee, S. H. Oh*. Geometry-driven surface electron dichotomy via ordered subsurface defects. Nature Physics (Under consideration).

2. K. Lee¶, Y. Xing¶, **J. Byun¶**, M. Jeong¶, J. Seo, J. Lee, J. Bang, E. Lee*, S. H. Oh*. Visualizing Oxygen Vacancy-Induced Dielectric Dead Layer at Metal-Oxide Interfaces by In-situ Multimodal Electron Microscopy. ***Science Advances*** (Under Review).

3. **J. Byun¶**, K. Lee¶, M. Jeong, E. Lee, J. Bang, H. Kim, G.H. Gu*, S.H. Oh*. Imaging 3D polarization dynamics via deep learning 4D-STEM. *arXiv preprint arXiv:2506.06598* (2025).

4. Z. Wang¶, **J. Byun¶**, Z. Wang, Y. Xing, J. Seo, J. Lee, S.H. Oh* Direct Observation of Atomic Step-Assisted Stabilization of Polar Surfaces. Advanced Materials, 35 (40), 2303051 (2023)

5. Z. Wang¶, **J. Byun¶**, S. Lee, J. Seo, B. Park, J. Kim, H.Y. Jeong, J. Bang, J. Lee, S.H. Oh* Vacancy driven surface disorder catalyzes anisotropic evaporation of polar ZnO (0001) surface. Nature Communications, 13, 5616 (2022)

6. Y.-I. Kim¶, M. Jeong¶, **J. Byun¶**, S.-H. Yang, W. Choi, W.-S. Jang, J. Jang, K. Lee, Y. Kim, J. Lee, E. Lee, Y.-M. Kim (2021). Atomic-scale identification of invisible cation vacancies at an oxide homointerface. Materials Today Physics, 16, 100302

7. J. Jo¶, **J. Byun¶**, J. Lee, D. Choe, I. Oh, J. Park, M.-J. Jin, J. Lee*, J.-W. Yoo* (2020). Emergence of Multispinterface and Antiferromagnetic Molecular Exchange Bias via Molecular Stacking on a Ferromagnetic Film. Advanced Functional Materials, 30 (11), 1908499

8. Y. Xing, I. Kim, K. T. Kang, **J. Byun**, W. S. Choi, J. Lee, S. H. Oh*. Monitoring the formation of infinite-layer transition metal oxides through in situ atomic-resolution electron microscopy. ***Nature Chemistry***, 17, 66–73 (2025).

9. J. Seo, H. Lee, K. Eom, **J. Byun**, T. Min, J. Lee, K. Lee, C.-B. Eom, S. H. Oh, Feld-induced modulation of two-dimensional electron gas at LaAlO3/SrTiO3 interface by polar distortion of LaAlO3. Nature Communications 15, 5268 (2024).

10. J. Hwang, **J. Byun**, H. Kim, J. Lee, High-mobility two-dimensional electron gas at the PbZr0.5Ti0.5O3/BaSnO3 heterostructure. J. Korean Phys. Soc. 84, 67–72 (2024).

11. Y. Ha, **J. Byun**, J. Lee, J. Son, Y. Kim, S. Lee, Infrared Transparent and Electromagnetic Shielding Correlated Metals via Lattice-Orbital-Charge Coupling. Nano Lett. 22, 6573–6579 (2022).

12. M. Sheen, Y. Ko, D. Kim, J. Kim, **J. Byun**, K. Y. Yeon, D. Kim, J. Jung, J. Choi, R. Kim, J. Yoo, I. Kim, C. Joo, N. Hong, J. Lee, S. H. Jeon, S. H. Oh, J. Lee, N. Ahn, C.Lee*. Highly efficient top-down processed blue InGaN nanoscale light-emitting diodes. *Nature*

13. K. Kang**, J. Byun**, M. Jeen, G. Jo, Y.K. Baek, J. Lee, H. Jeen (2022). Effect of sintering time on electronic properties of strontium hexaferrite. *Ceramics International*

14. K. Lee, **J. Byun**, K. Park, S. Kang, M.-S. Song, J. Park, J. Lee, S. Chae* (2022). Giant Tunneling Electroresistance in Epitaxial Ferroelectric Ultrathin Films Directly Integrated on Si. *Applied Materials Today, 26,* 101308

15. J. Lee, M. S. Song, W.-S. Jang, **J. Byun**, H. Lee, M. H. Park, J Lee, Y-M Kim*, S. C. Chae*, T. Choi* (2021) Modulating the ferroelectricity of hafnium zirconium oxide ultrathin films via interface engineering to control the oxygen vacancy distribution, *Advanced Materials Interfaces, 9 (7),* 2101647

16. Y. Ha, **J. Byun**, J. Lee, S. Lee (2021). Design Principles for the Enhanced Transparency Range of Correlated Transparent Conductors. *Laser & Photonics Reviews*, 15 (5), 2000444

17. J. Jo[¶], **J. Byun**, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J. Lee*, J.-W. Yoo* (2018). Molecular tunability of magnetic exchange bias and asymmetrical magnetotransport in metalloporphyrin/Co hybrid bilayers. *ACS Nano*, 13 (1), 894-903

18. S.-J. Kwon, B.-M. Jung, T. Kim, **J. Byun**, J. Lee, S.-B. Lee, U.-H. Choi * (2018). Influence of Al_2O_3 Nanowires on Ion Transport in Nanocomposite Solid Polymer Electrolytes. *Macromolecules*, 51(24), 10194-10201

19. T.T. Ly, G. Duvjir, T. Min, **J. Byun**, T. Kim, M.M. Saad, N.T.M. Hai, S. Cho, J. Lee*, J. Kim* (2017). Atomistic study of the alloying behavior of crystalline $SnSe_{1-x}S_x$. *Physical Chemistry Chemical Physics*, 19 (32), 21648-21654

CONFERENCE PRESENTATIONS

TALKS

1. **J. Byun**, Z. Wang, S. Oh, J. Lee. (2021) Novel Stabilization Mechanism on Polar Oxide Surface. Talk presented at 2021 Materials Research Society Fall Meeting

2. **J. Byun**, Z. Wang, S. Oh, J. Lee. (2021) Novel Stabilization Mechanism on Polar Oxide Surface. Talk presented at 2021 Korean Physics Society Fall Meeting

3. **J. Byun**, J. Lee. (2021). Giant Nonlinear Tunneling Current in HfO_2-based Anti-Ferroelectric Tunnel Junction. Talk presented at the Electronic Materials and Applications Conference 2021.

4. **J. Byun,** J. Lee, T. Choi, J. Lee. (2021). Interface engineered TER optimization in HfO_2-based FTJ. Talk presented at the 2021 Korean Physics Society Spring Meeting

5. **J. Byun,** J. Lee, T. Choi, J. Lee. (2020). DFT investigation on structural stability on the ferroelectric HfO_2 on the electrode. Talk presented at the 2020 Korean Physics Society Fall Meeting

6. **J. Byun**, T. Min, J. Lee. (2020). Nonlinear current-voltage behavior in HfO_2-based ferroelectric domain-wall tunnel junction. Talk presented at the 2020 Korean Ceramic Society Spring Meeting

7. **J. Byun**, T. Min, J. Lee. (2020). Giant Nonlinear Tunneling Current in HfO_2-based Anti-Ferroelectric Tunnel Junction. Talk presented at the America Physics Society March Meeting 2020

8. **J. Byun**, T. Min, J. Lee. (2019). First-principles studies of the effects of oxygen vacancies on the HfO_2-based ferroelectric tunnel junction. Talk presented at the 2019 Materials Research Society Fall Meeting

9. **J. Byun**, T. Min, J. Lee. (2019). First-principles studies of the HfO_2-based ferroelectric tunnel junction. Talk presented at the 5th International Conference on Advanced Electromaterials

10. **J. Byun**, T. Min, J. Lee. (2019). Design and modeling of optimal HfO_2-based ferroelectric tunneling junction. Talk presented at the 2019 Korean Physics Society Fall Meeting

11. **J. Byun**, T. Min, J. Lee. (2019). Design and modeling of optimal HfO_2-based ferroelectric tunneling junction. Talk presented at the Asia-Pacific PFM 2019 Conference

12. **J. Byun**, T. Min, J. Lee. (2019). Complex bandstructure calculation for the ferroelectric tunneling junction current. Talk presented at the 2019 Korean Physics Society Spring Meeting

13. **J. Byun**, T. Min, J. Lee. (2018). First-principles studies of the effects of effective mass on the HfO_2-based ferroelectric tunnel junction. Talk presented at the 2018 Korean Physics Society Fall Meeting

14. **J. Byun**, T. Min, J. Lee. (2018). Novel high coercivity $Fe_{16-x}Al_xN_2$ alloy. Talk presented at the IcAUMS 2018

15. **J. Byun**, T. Min, J. Lee. (2018). High coercivity $Fe_{16-x}Al_xN_2$ alloy design. Talk presented at the Korean Magnetics Society 2018 Spring Conference

16. **J. Byun**, T. Min, J. Lee. (2018). Novel high coercivity $Fe_{16-x}Al_xN_2$ alloy. Talk presented at the American Physics Society March Meeting 2018

17. **J. Byun**, T. Min, J. Lee. (2017). First-principles modeling of electron tunneling through ferroelectric tunnel junction. Talk presented at the 2017 KPS Fall Meeting

18. **J. Byun**, T. Min, J. Lee. (2017). Design of $Fe_{16-x}Al_xN_2$ with high-coercivity. Talk presented at the 2017 KPS Spring Meeting

POSTERS

1. **J. Byun**, J. Jo, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J.-W. Y, J. Lee. (2019). First principles study of magnetic exchange interaction in Metalloporphyrin/Co interface. Poster presented at the 3rd Workshop on Functional Materials Science.

2. **J. Byun**, T. Min, J. Lee. (2019). First-principles studies of the HfO_2-based ferroelectric tunnel junction. Poster presented at the 15th KIAS Electronic Structure Calculation Workshop

3. **J. Byun**, T. Min, J. Lee. (2019). Ab initio study of the role of effective mass in ferroelectric tunneling junction. Poster presented at the The 15th Dielectric Union Symposium

4. **J. Byun**, T. Min, J. Lee. (2019). First-principles studies of the effects of oxygen vacancy and effective mass on the HfO_2-based ferroelectric tunnel junction. Poster presented at the 10th APCTP WORKSHOP ON MULTIFERROICS.

5. **J. Byun**, T. Min, J. Lee. (2018). First-principles studies of the effects of oxygen vacancies on the HfO_2-based ferroelectric tunnel junction. Poster presented at the 12th Japan-Korea Conference on Ferroelectrics

TECHNICAL SKILLS

ML code: Pytorch
DFT/MD code: VASP, QuantumEspresso, CP2K, Wannier90, Phonopy, GULP
Programing: Python, C++(Arduino)
Scientific computing environment: Linux server management, Queuing system(SGE), Git

OTHER

2013-2014 **Military service**

2017	**Python study manager for graduate students**
2018	**KIAS CAC Summer School on Scientific Computing and Machine Learning**
2021-2021	**The president of the student council**

GITHUB

https://github.com/material-JH

RESEARCH INTEREST

Research Tools: **Deep Learning, Ab initio multislice simulation**, *Ab initio* MD(Molecular Dynamics), MLFF(Machine Learning Force Field), Phonon property
Materials: Oxide Surface and interface, Perovskite, 2D materials